

From Spectral Interference to Overlapped Features: A Full-Spectrum LIBS Voigt Decomposition Strategy with Shoulder Detection and Residual Completion

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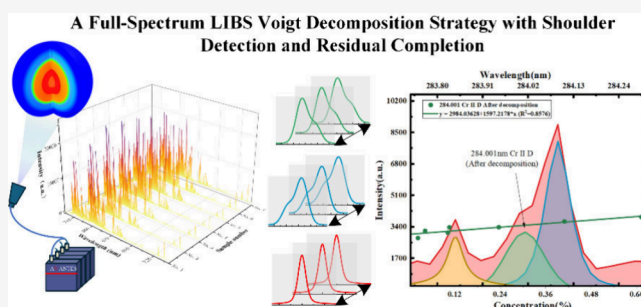
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ABSTRACT: Laser-induced breakdown spectroscopy (LIBS) provides rapid, multielement analysis of complex alloys, but dense line crowding, limited resolution, and plasma broadening cause strong spectral interference and overlapping features that degrade quantitative performance. We propose a full-spectrum Voigt decomposition strategy for LIBS that combines shoulder detection with residual-based line completion. The measured spectrum is first segmented into interference-aware regions of interest (ROIs) using line lists and instrumental resolution. Within each ROI, candidate peaks and shoulders are automatically identified and used as seeds for constrained multicomponent Voigt (or pseudo-Voigt) fitting, with bounds on line width, center drift, and amplitude derived from instrument and plasma conditions. The resulting residual spectrum is then treated as informative: localized residual structures are projected onto Voigt-shaped templates, and their amplitudes are tested statistically to decide whether additional components are required, thereby recovering missed weak or shoulder-like lines. Tests on synthetic spectra constructed from database lines and on experimental LIBS spectra of steels show improved separation of overlapped features, reduced relative standard deviation of line intensities, and enhanced correlations with elemental concentrations. The method provides a scalable, physically grounded framework for decomposing interference- and overlap-affected LIBS spectra.



INTRODUCTION

Laser-induced breakdown spectroscopy (LIBS) has become an important platform for rapid, in situ elemental analysis in materials science, environmental monitoring and planetary exploration.^{1–3} A single laser pulse generates a microplasma that emits a dense forest of atomic and ionic lines, enabling simultaneous detection of major and minor elements with minimal sample preparation.^{4,5} This is particularly attractive for multicomponent alloys such as stainless steels, where online monitoring of Cr, Ni, Mn and other alloying elements is increasingly required.^{6,7} Yet the same spectral richness that makes LIBS powerful also makes it difficult to interpret: strong spectral interference, finite instrumental resolution and plasma-induced broadening jointly produce complex, heavily overlapped spectra, in which weak lines and shoulder structures are easily masked and apparent intensities are biased by neighboring transitions and background structure.^{8,9}

To alleviate these limitations, a first class of approaches operates at the “global spectrum” level. Spectrum standardization converts measured line intensities to a virtual standard plasma state (fixed temperature, degree of ionization and number density), thereby mitigating shot-to-shot fluctuations and matrix effects before calibration.^{10,11} Calibration-free

LIBS (CF-LIBS) goes further by inferring plasma parameters and elemental concentrations directly from the spectrum; recent reviews document substantial progress through improved plasma models, self-absorption corrections and optimization strategies.^{12,13} These methods have clearly improved robustness and reduced dependence on matrix-matched standards, but they usually assume that the underlying spectral features have already been correctly isolated or at least not severely distorted by unresolved overlap. In practice, any residual interference at the feature level—including unrecognized shoulders and weak overlapped lines—is propagated into the standardized or CF-LIBS workflow.

A second line of research deals with spectral interference more explicitly at the level of individual peaks. Curve-fitting methods describe each emission line with Gaussian, Lorentzian

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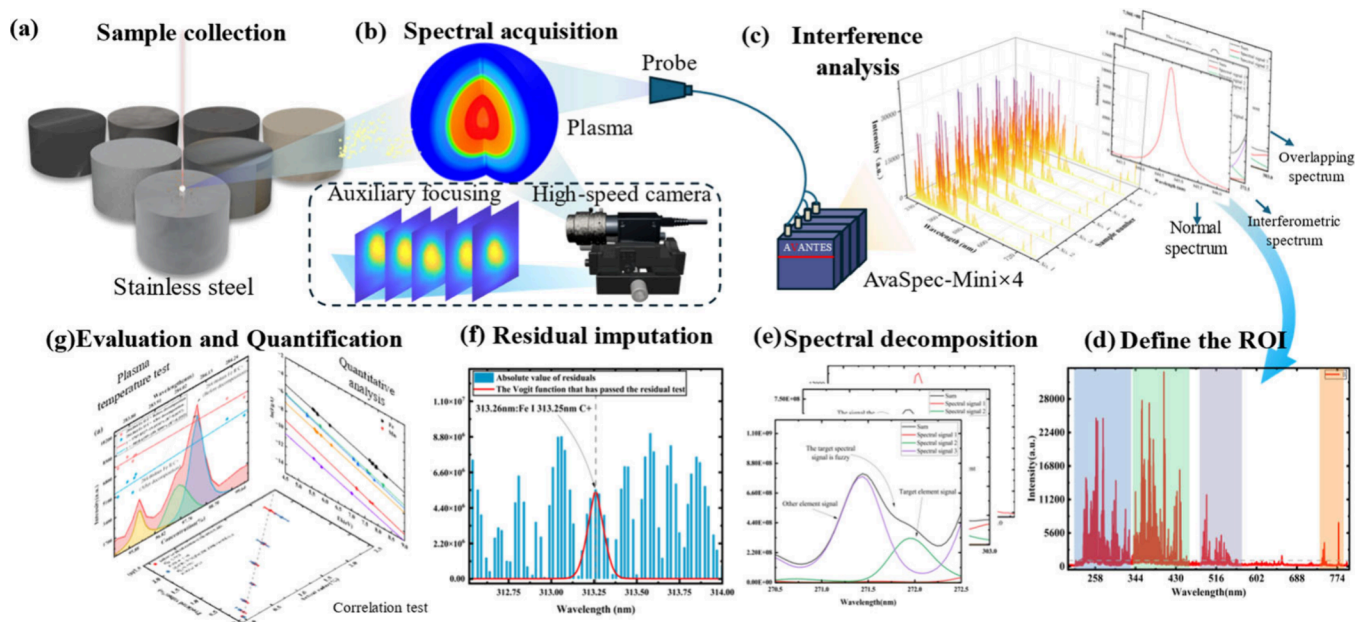


Figure 1. Workflow of LIBS spectral acquisition, decomposition, and residual-based line imputation for stainless-steel analysis: (a) sample collection, (b) spectral acquisition, (c) interference analysis, (d) defining the ROI, (e) spectral decomposition, (f) residual imputation, and (g) evaluation and quantification.

or Voigt profiles and use multipoint fitting to decompose overlapping features and correct intensities before quantitative analysis.^{14,15} For example, Tan¹⁶ et al. demonstrated decomposition and error compensation of overlapping LIBS peaks using multipoint Voigt fitting, improving the accuracy of interfered lines. Variable-selection strategies combined with multivariate analysis have been proposed as an alternative to multipoint fitting for spectra with overlapping lines, effectively redistributing interference within a chemometric model rather than resolving it explicitly. More recently, automatic or unsupervised processing pipelines have been developed: Labutin¹⁷ et al. introduced an algorithm for automatic identification of emission lines in LIBS spectra by correlating experimental spectra with synthetic line databases, and Ewusi-Annan and Melikechi¹⁸ presented an unsupervised fitting scheme that scales emission-line fitting to large LIBS data sets. Collectively, these studies show that physics-based line-shape modeling and automation can substantially improve the treatment of spectral interference, but they often rely on manually defined regions of interest (ROIs), do not explicitly target shoulder features, and rarely exploit the residual spectrum as a source of additional information.

Despite recent advances, constructing a scalable decomposition workflow for dense LIBS spectra remains difficult.^{19,20} In many studies, ROIs are defined manually, component numbers per ROI are selected heuristically, and fit quality is judged mainly from R^2 or visual inspection, so the envelope is reproduced but weak or shoulder-like Voigt components may remain unresolved, and residuals may contain structured, nonrandom features rarely subjected to explicit statistical tests.

A full-spectrum LIBS framework is developed that combines ROI-based seeding, shoulder detection and residual-aware refinement within a constrained multi-Voigt decomposition. The spectrum is partitioned into interference-aware ROIs using line lists; within each ROI, candidate peaks and shoulders are detected automatically and used as seeds for multicomponent Voigt fitting, and the resulting residuals are

treated as a secondary signal whose localized structures are matched to Voigt templates and tested for statistical significance to decide whether additional components are required. Applied to steel LIBS spectra, this shoulder-guided, residual-refined decomposition yields deconvoluted line sets with reduced relative standard deviations of line intensities and improved correlations with elemental concentrations, providing cleaner features for both calibration-based and calibration-free quantification.

METHOD

As illustrated in Figure 1, a complete processing workflow is established from sample excitation to quantitative spectral evaluation. In Figure 1(a), a cylindrical stainless-steel sample is irradiated by the laser to generate a plasma plume under reproducible excitation conditions. In Figure 1(b), the plasma emission is collected by an optical probe and auxiliary focusing optics, while a high-speed camera records its spatiotemporal evolution. In Figure 1(c), the original spectra and interference-affected spectra are synchronously acquired by a multichannel spectrometer to identify overlapping and interfering emission lines. On this basis, Figure 1(d) shows the partitioning of the full spectrum into wavelength bands and the definition of regions of interest (ROIs), restricting subsequent fitting to intervals where the target features are dominant. In Figure 1(e), the first derivative and a shoulder-search strategy are applied within each ROI to locate decomposition points, followed by multicomponent Voigt fitting to extract the line center, broadening and intensity parameters. In Figure 1(f), the residual sequence is subjected to statistical analysis and residual-based interpolation to identify potentially missed weak peaks. Finally, in Figure 1(g), correlation analysis and plasma temperature tests are combined to complete the quantitative evaluation.

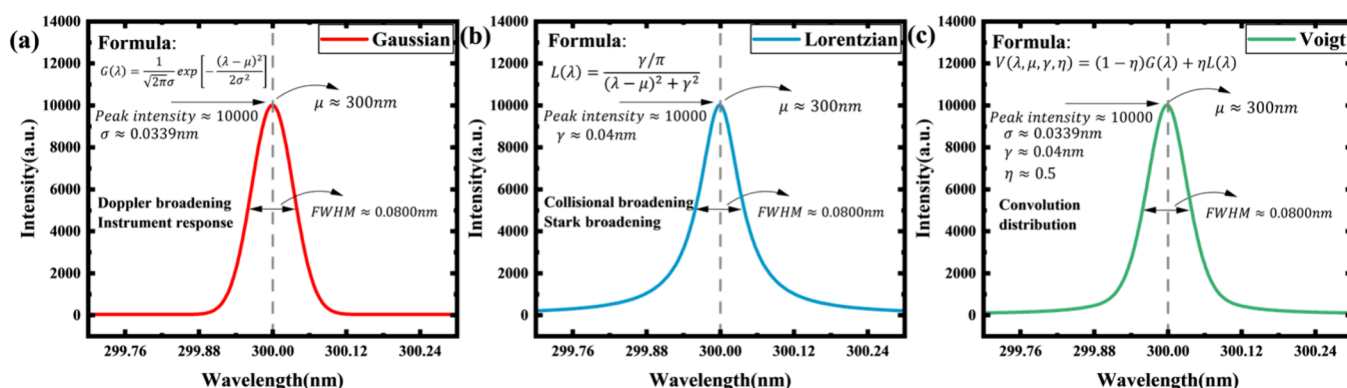


Figure 2. (a) Gaussian spectrum. (b) Lorentzian spectrum. (c) Voigt spectrum.

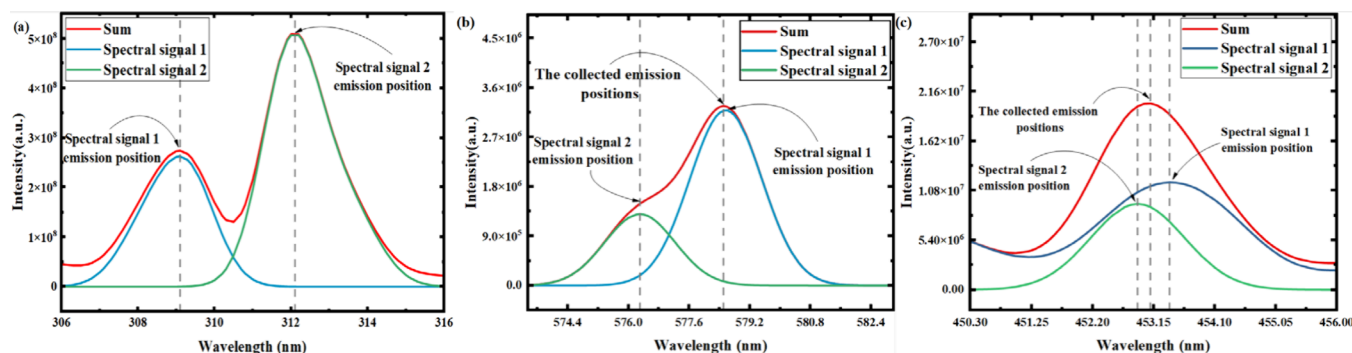


Figure 3. (a) Spectral interference. (b) Spectral overlap. (c) Overlapped peaks.

Theoretical Background and Motivation

In LIBS, each emission line corresponds to a radiative transition between two electronic energy levels of an atom or ion. Its central wavelength is determined by the energy difference between the upper and lower levels.²¹ In an idealized case, a transition with energy gap $E_u - E_l$ would emit at a single, exact frequency. In reality, the energy levels are not infinitely sharp. The finite lifetime of the excited state, thermal motion, collisions, external fields and other perturbations all broaden the line. As a result, the emission profile changes from a δ -function to a finite-width distribution, typically described by Gaussian, Lorentzian or Voigt functions.²² When line broadening is dominated by Doppler effects or by finite instrumental resolution, the velocity distribution of thermally moving atoms or ions produces a characteristic Gaussian profile, as illustrated in Figure 2(a).²³ If the broadening is mainly caused by collisional or Stark effects, the finite level lifetime and microfield-induced splitting lead, via the energy–time uncertainty principle, to a Lorentzian line shape, as shown in Figure 2(b).²⁴ In most practical LIBS measurements, Doppler and collisional broadening coexist. Spectral lines are simultaneously affected by thermal motion and by plasma microfields, and are further convolved with the instrument response function. When Gaussian and Lorentzian mechanisms act together, the resulting theoretical line profile takes the Voigt form, as depicted in Figure 2(c), representing a composite of the two broadening contributions.²⁵

From the above discussion, it follows that the essence of spectral overlap is the mutual superposition of the energy distributions of finitely broadened line profiles. When the separation between two transitions is smaller than their effective line widths, their profiles merge into an overlapped

peak. In LIBS plasmas, the high temperature and high particle density greatly increase the probability of such overlap. According to the Rayleigh criterion, a lower instrumental resolving power further promotes overlap, since lines can only be distinguished when their separation exceeds the resolution limit. This is illustrated in Figure 3(a), which shows spectral interference, Figure 3(b), where distinct lines begin to overlap, and Figure 3(c), where they form a fully merged peak. Spectral overlap is therefore one of the most common and challenging problems in quantitative LIBS: it not only distorts the apparent line shape, but can also trigger misidentification, systematic errors in line intensities, and propagated biases in temperature and concentration retrieval. As a result, spectral decomposition becomes a necessary means of recovering the underlying line information.

In summary, line overlap is a major factor limiting both the quantitative accuracy and the reliability of physical diagnostics in LIBS. Extracting the true contribution of each transition from a complex superposition cannot be achieved by improving instrumental resolution alone. Spectral decomposition techniques model and fit the measured signal, allowing different component peaks to be distinguished even under finite resolution and recovering the independent contribution of each physical parameter. In doing so, they can significantly enhance the precision of spectral interpretation and the fidelity of plasma parameter inversion.

ROI Partitioning

In the acquired spectra, the high abundance of alloying elements in steel gives rise to thousands of emission lines between 200 and 700 nm. These lines are densely distributed with widely varying intensities, and neighboring lines are often broadened or partially overlapped. If one attempts to perform a

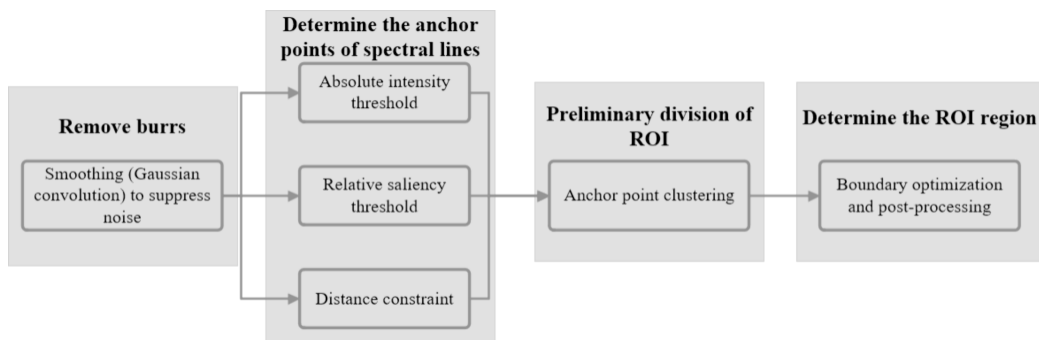


Figure 4. Flowchart of the ROI segmentation process.

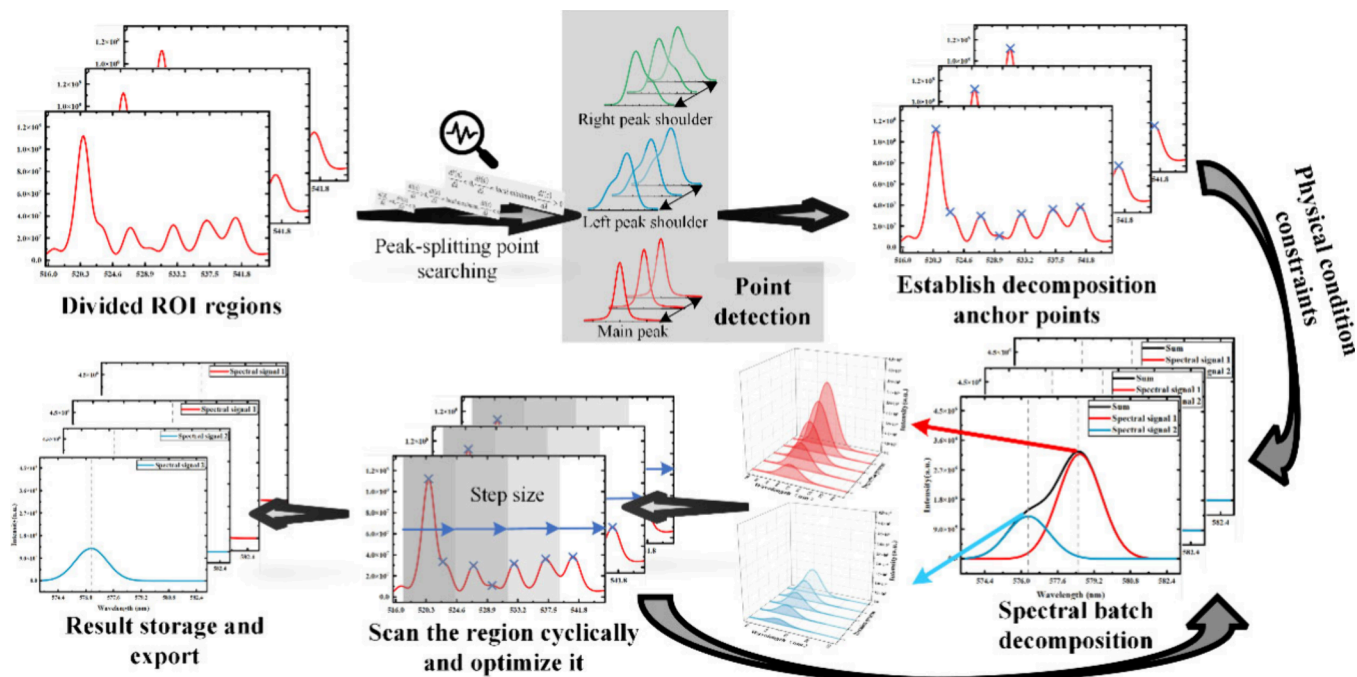


Figure 5. Flowchart of the pseudo-Voigt decomposition process.

global fit over the entire spectrum, the computation becomes extremely expensive and the model suffers from an excessive number of parameters, strong coupling between adjacent peaks, poor convergence behavior and a high risk of being trapped in local minimum. It is therefore necessary to first localize the analysis by dividing the spectrum into regions of interest (ROIs). The key rationale for defining ROIs is to ensure that, within each region, the included spectral lines share similar physical origins and line-shape characteristics.

The basic principle of ROI segmentation is illustrated in Figure 4. Each ROI contains only a limited number of spectral lines; within this interval the background varies slowly and the broadening mechanism can be regarded as uniform. Consequently, the dimensionality of the optimization problem is significantly reduced, which helps avoid the ill-posed nature of global fitting and lowers computational complexity, while ensuring that the local fitting remains physically interpretable. First, candidate peak positions are determined from the local extremum structure of the intensity function $I(\lambda)$. Local maxima are identified using the first- and second-derivative conditions:

$$\begin{cases} \frac{dI(\lambda)}{d\lambda} = 0 \\ \frac{d^2I(\lambda)}{d\lambda^2} < 0 \end{cases} \quad (1)$$

To suppress noise artifacts, a smoothing kernel $G(\lambda)$ is introduced to generate the smoothed signal I_S via the convolution $I_S = I * G$. Based on this smoothed profile, candidate features are screened using an absolute intensity threshold I_{abs} and a relative significance threshold T_{rel} . The selection criteria are defined as follows:

$$I(\lambda) > I_{abs}, I(\lambda) - \max(I_L, I_R) \geq T_{rel} \quad (2)$$

Peaks satisfying the above criteria are designated as anchor points. Adjacent anchors are then clustered based on wavelength spacing: if the distance between two peaks is less than a threshold $\varepsilon = \alpha * fwhm$, they are grouped into a single peak cluster. Here, α represents a coefficient governing the clustering scale. The initial left and right boundaries of each cluster are defined by the adjacent local minima. However, if the spectral intensity at these boundaries remains above a specified threshold τ , an energy integration criterion is applied:

Where α represents a coefficient governing the clustering scale. The initial left and right boundaries of each cluster are defined by the adjacent local minima. However, local minima may not always fully enclose the spectral contribution of a peak cluster, especially in dense or overlapping spectral regions. To ensure that the complete spectral energy is preserved, an energy-based criterion is introduced. The spectral energy within a candidate region $[a, b]$ is defined as

$$E(a, b) = \int_a^b \max(I(\lambda) - I_{\text{base}}, 0) d\lambda \quad (3)$$

A window expansion is subsequently performed, but it is not unconditional. For each provisional region, the algorithm evaluates an edge-energy criterion on both boundaries using the energy integral defined in eq 3 with a locally estimated baseline. The boundary is expanded in discrete steps only when the edge energy exceeds a prescribed fraction of the adjacent inner-band energy, and the expansion terminates once this criterion is no longer satisfied or a maximum number of steps is reached. After expansion, refinement rules are applied. If a region width exceeds the maximum allowable limit W_{max} , the region is split with priority given to local minima in the smoothed spectrum; if no qualified minimum exists, an equal-width split is used as a fallback. Conversely, if the gap between adjacent regions is smaller than the threshold δ , the two regions are merged. Through these steps, the spectrum is adaptively segmented into discrete ROIs, providing physically interpretable subregions that serve as stable inputs for the subsequent multippeak decomposition and parameter retrieval.

Voigt Decomposition Strategy

Traditional Voigt fitting typically applies global modeling across the full spectrum or within fixed windows. However, this approach involves a large number of parameters, depends strongly on initial peak position estimates, and lacks physical constraints, which frequently leads to unstable decomposition of heavily overlapped features. Formally, standard Voigt fitting decomposes the spectrum within a given window using the Voigt formula:

$$V(\lambda; \mu; \sigma; \gamma) = \int_{-\infty}^{+\infty} G(\lambda'; \mu, \sigma) L(\lambda - \lambda'; \mu, \gamma) d\lambda' \quad (4)$$

However, the numerical solution of the exact Voigt integral is computationally intensive. To address this, we propose an improved pseudo-Voigt decomposition method that incorporates adaptive ROI segmentation and physical constraints, thereby enhancing the speed, robustness, and physical interpretability of the decomposition process. The overall workflow is illustrated in Figure 5.

Within each ROI, a search for peak-splitting points and feature detection are performed to identify main peaks and their associated left and right shoulders, which serve as potential anchors for spectral decomposition. Once these anchors are established, the algorithm executes a batch spectral decomposition subject to physical constraints. Each subpeak is then iteratively fitted using cyclic scanning and step-size optimization. Finally, the decomposed spectral parameters are output and saved. This workflow achieves a closed-loop optimization involving spectral segmentation, peak detection, physically constrained fitting, and result export, demonstrating a fully automated and interpretable spectral decomposition strategy.

First, a physically constrained seed-peak identification mechanism is employed within each ROI. The positions of the main peaks are detected based on the first-derivative extremum criteria, as expressed by

$$\frac{dI(\lambda)}{d\lambda} = 0, \frac{d^2I(\lambda)}{d\lambda^2} < 0, I(\lambda) \geq k_{\text{seed}} I_{\text{base}} \quad (5)$$

Here, I_{seed} denotes the baseline intensity, k_{seed} represents the baseline amplification factor, ensuring that the identified main peak is significantly distinguishable from the background. Additionally, a derivative trend analysis is introduced to detect subtle shoulders that are not visually obvious. Specifically, for a right-side shoulder, if the first derivative $dI(\lambda)/d\lambda$ exhibits a "decrease-minimum-increase" pattern—that is, there exist points $a < b < c$ such that

$$\frac{dI(a)}{d\lambda} < 0, \frac{dI(b)}{d\lambda} = \text{local minimum}, \frac{dI(c)}{d\lambda} > 0, \\ I(b) \geq k_{\text{seed}} I_{\text{base}} \quad (6)$$

The *local minimum* denotes the local minimum value within the neighborhood (a, c) , which indicates the presence of a right-side shoulder in the vicinity. Conversely, for a left-side shoulder, if the first derivative $dI(\lambda)/d\lambda$ exhibits an "increase-maximum-decrease" trend—implying there exist $a < b < c$ such that

$$\frac{dI(a)}{d\lambda} > 0, \frac{dI(b)}{d\lambda} = \text{local maximum}, \frac{dI(c)}{d\lambda} < 0, \\ I(b) \geq k_{\text{seed}} I_{\text{base}} \quad (7)$$

The *local maximum* denotes the local maximum value within the neighborhood (a, c) . To prevent false detections triggered by noise, a consistency constraint is imposed requiring the derivative to maintain a uniform sign within a local window. Subsequently, in the fitting phase, the observed spectrum within each ROI is modeled as a non-negative linear superposition of multiple pseudo-Voigt components:

$$I_k(\lambda) = \sum_{i=1}^{N_k} A_i P(\lambda; \mu_i, \sigma_i, \gamma_i, \eta_i) \quad (8)$$

Where P denotes the pseudo-Voigt components, with its mathematical expression given by

$$P(\lambda; \mu_i, \sigma_i, \gamma_i, \eta_i) = \eta_i L(\lambda; \mu_i, \gamma_i) + (1 - \eta_i) G(\lambda; \mu_i, \sigma_i) \quad (9)$$

Where η_i ($0 \leq \eta_i \leq 1$) represents the Lorentzian fraction. Unlike fixed-parameter approaches, η_i is treated as an optimizable variable in our model. It is initialized at 0.5 and subsequently refined via residual minimization within each ROI. This allows the model to adaptively capture the variation between Gaussian-dominated and Lorentzian-dominated profiles without manual intervention. $G(\lambda)$ and $L(\lambda)$ denote the normalized Gaussian and Lorentzian profiles. A non-negative least-squares constraint is applied to the intensity A_i of each component:

$$\min_{A_i \geq 0} \|I_{\text{obs}} - XA\|^2 \quad (10)$$

This ensures physical non-negativity and mitigates overfitting. Simultaneously, the optimization objective is to minimize the residual:

$$\min_{\theta} J(\theta) = \sum_{\lambda} \left[I_{\text{obs}}(\lambda) - \sum_{i=1}^{N_k} A_i P(\lambda; \mu_i, \sigma_i, \gamma_i, \eta_i) \right]^2 \quad (11)$$

Finally, the change in goodness of fit upon removing the j -th component is calculated as

$$\Delta R_j^2 = R_{\text{full}}^2 - R_{(-j)}^2 \quad (12)$$

This metric quantifies the independent contribution of each spectral line, allowing for the elimination of spurious components and the assessment of fitting reliability. Furthermore, by determining accurate initial peak positions, the proposed scheme establishes a physically reasonable starting point and leverages the monotonic decrease of the error function to ensure automatic convergence to the optimal solution.

EXPERIMENTAL SETUP AND DATA ACQUISITION

Experimental Setup

The experimental LIBS setup is illustrated in Figure 6. A Q-switched Nd:YAG laser (Ultra 100, Quantel) operating at a fundamental

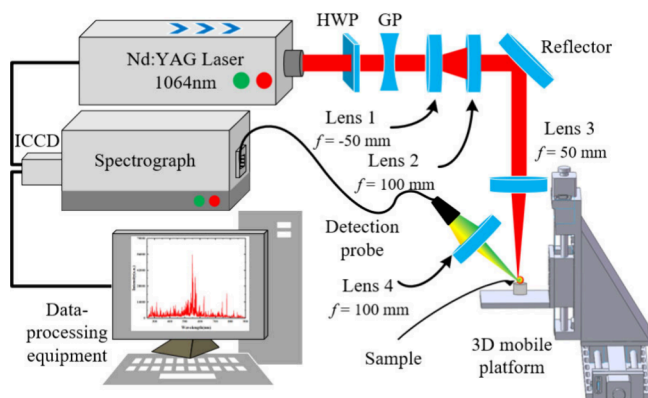


Figure 6. Schematic of the experimental setup.

wavelength of 1064 nm was employed as the excitation source, featuring a pulse duration of approximately 10 ns and a repetition rate of 1 Hz. The laser pulse energy was adjusted using a combination of a half-wave plate (HWP) and a Glan prism (GP). The laser beam was first expanded by a telescope system consisting of a concave lens L1 ($f = -50$ mm) and a convex lens L2 ($f = 100$ mm). Subsequently, the beam was focused onto the sample surface by a plano-convex lens L3 ($f = 50$ mm) to induce the plasma. The plasma emission was collected at an angle of 45° relative to the incident beam by a convex lens L4 ($f = 100$ mm) and coupled into an optical fiber connected to a four-channel spectrometer (AvaSpec-Mini4096CL, Avantes, Netherlands). The spectrometer covers a spectral range of 186–797 nm with a resolution of approximately 0.05 nm and a minimum gate width of 9

μ s. For the experimental acquisitions, the delay time and gate width were set to 1.5 and 9 μ s, respectively.

Data Acquisition

Seven steel alloy samples were selected for this study, provided by the National Center for Iron and Steel Materials Testing and the Shandong Institute of Metallurgy. The elemental compositions of these samples were certified using Inductively Coupled Plasma Atomic Emission Spectroscopy (ICP-AES), as listed in Table 1, with the matrix element (Fe) content exceeding 90% in all cases. To mitigate the influence of surface contamination on the analysis and to enhance detection efficiency, five distinct sampling spots were selected on each sample. At each sampling point, single-shot spectra were screened for stability and 50 stable spectra were retained for subsequent analysis. The screening was based on the consistency of the overall emission level and spectral-shape similarity, and unstable shots were identified as outliers using robust statistics. The thresholds were chosen empirically to exclude evidently unstable pulses. Consequently, a total of 1750 ($50 \times 5 \times 7$) spectral data sets were synchronously acquired for each beam type.

RESULTS AND DISCUSSION

Decomposition Results

To validate the effectiveness of the proposed decomposition algorithm, representative synthetic mixture spectra were generated using the NIST LIBS database. As illustrated in Figure 7, the simulation parameters were configured as follows: a composition of Fe (90%), Ni (5%), and Cr (5%); an electron temperature of 1 eV; an electron density of $1 \times 10^{17} \text{ cm}^{-3}$; and a resolution of 2000. Two spectral intervals exhibiting common interference patterns were selected for decomposition analysis: 312.5–314.9 nm and 325–327.5 nm. The corresponding spectra are presented in Figure 7(a) and 7(d), respectively.

As observed in Figure 7(b), for the 325–327.5 nm region where the spectrum is dense, but interference is moderate, the intensity recovery rate for Fe I reached 99%. The line width estimation accuracy within this interval ranged from a maximum of 99.3% (predicted 0.302 nm vs actual 0.3 nm) to a minimum of 80.5% (predicted 0.251 nm vs actual 0.3 nm). Peak positions were accurately aligned with no shifting, and the goodness of fit (R^2) reached 0.9627, indicating an excellent decomposition. The residuals of individual components are shown in Figure 7(c). Larger residuals are observed on both sides of 325.92 nm; this resulted from the fitting of the Fe II line, where the presence of neighboring peaks on the left caused a conservative estimation of the broadening, leading to deviations at the wings. Apart from this, the fitting performance for other signals in this region was effective with minimal residuals. In contrast, when applied to the complex mixed signal in the 312.5–314.9 nm region, the decomposer successfully separated signals but with reduced performance. Since Fe I lacked a distinct emission feature in this heavily interfered region, the algorithm failed to identify it as an initial

Table 1. Certified Concentrations of Steel Alloy Samples (wt.%)

Sample	Si	Mn	Cr	Ni	Cu	Als	Ti	V	Fe
1	0.054	0.018	0.023	0.011	0.0036	0.0069	0.0002	0.0001	99.7784
2	0.077	0.104	0.042	0.045	0.049	0.024	0.02	0.011	99.1698
3	1.55	0.303	0.236	0.563	0.403	0.295	0.049	0.052	95.8989
4	1.09	0.669	0.102	0.4	0.316	0.214	0.202	0.098	95.9517
5	0.769	1.04	0.106	0.393	0.248	0.101	0.178	0.257	95.6516
6	0.436	1.47	0.409	0.206	0.167	0.05	0.124	0.201	95.8404
7	0.176	2.1	0.612	0.107	0.075	0.022	0.088	0.147	95.3222

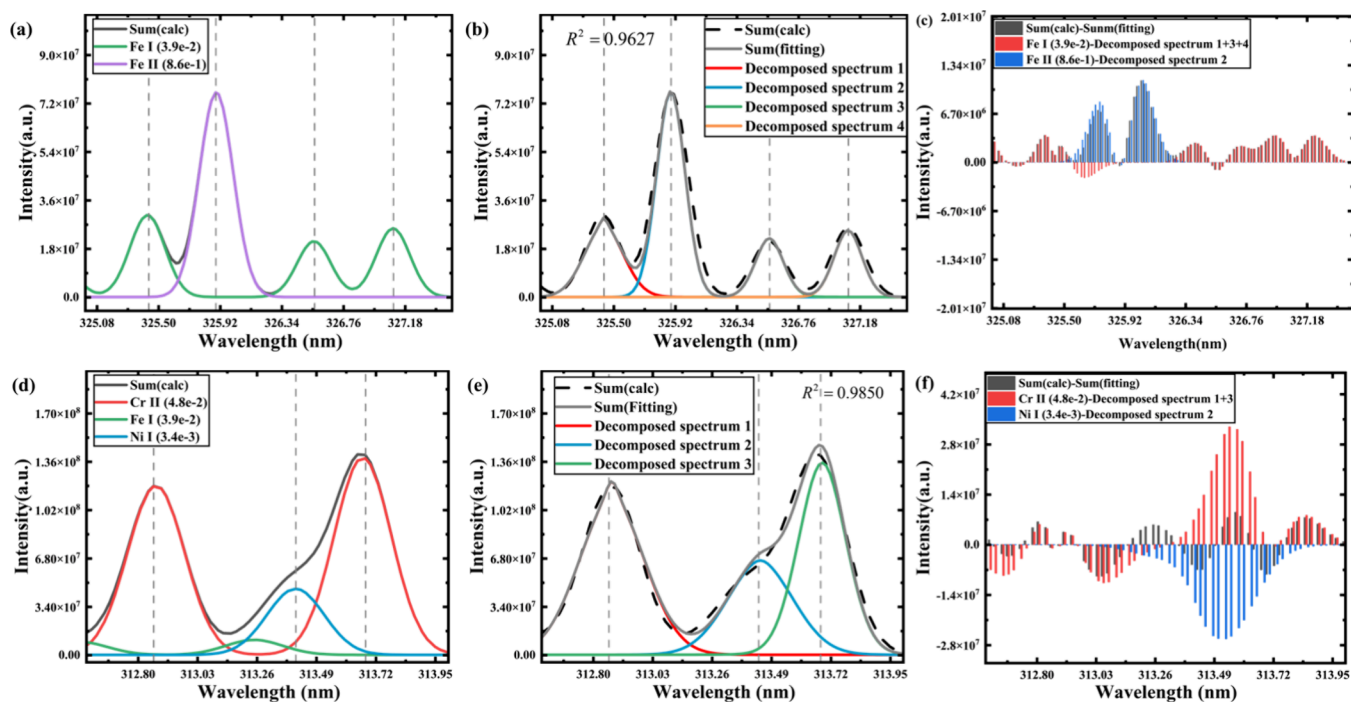


Figure 7. Synthetic spectrum generated using the NIST LIBS database. For the 325–327.5 nm range: (a) original spectrum, (b) decomposition results, and (c) residual comparison between synthetic and decomposed components. For the 312.5–314.9 nm range: (d) original spectrum, (e) decomposition results, and (f) residual comparison between synthetic and decomposed components.

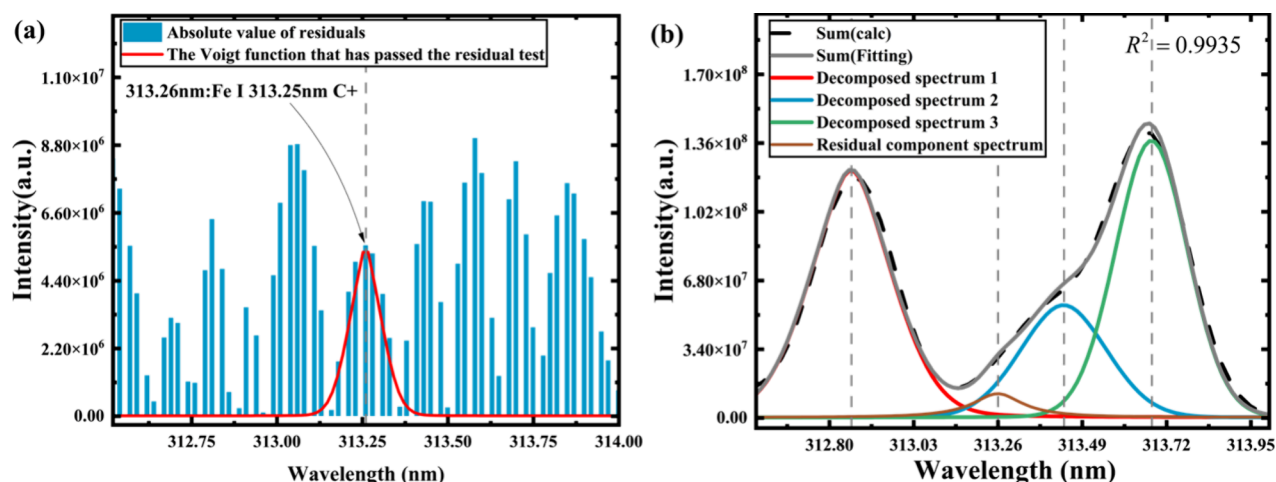


Figure 8. (a) Residual completion analysis and (b) refined decomposition results after completion.

seed, locating only three signals. Consequently, the decomposition yielded goodness-of-fit values of 0.9981 for Cr II and 0.5300 for Ni I. While the Cr II position was accurate, the Ni I position shifted by 0.03 nm with a significant amplitude deviation, resulting in a poor fit. To further optimize the decomposition, a residual completion mechanism was introduced. Residual completion is used to compensate underfitting in spectral decomposition by completing structured residuals after the initial ROI fitting. This method detects whether the remaining residuals conform to a Voigt distribution; if so, an additional decomposition point is permitted. To prevent overfitting through unrestricted component addition, the position of the maximum residual is cross-referenced with the NIST Atomic Spectra Database. A new decomposition point is generated only if a corresponding emission line exists in the database. Ultimately, a new

decomposition position was identified at 313.26 nm within the interval. The model was then allowed to readjust the existing components, and the refined results are presented in Figure 8.

Taking the absolute value of the residuals in the 312.5–314.9 nm range yields the data set presented in Figure 8(a). Visual inspection reveals that the structure centered at 313.25 nm exhibits a symmetric distribution consistent with a Voigt function. In contrast, residual structures at other positions lack symmetry, suggesting they represent imperfect broadening compensation (residuals from “conservative” fitting) rather than missed spectral lines; consequently, Voigt fitting parameters fail to converge in these areas. To objectively verify the nature of the residuals in this ROI, a hypothesis testing framework using a *t* test was established to distinguish

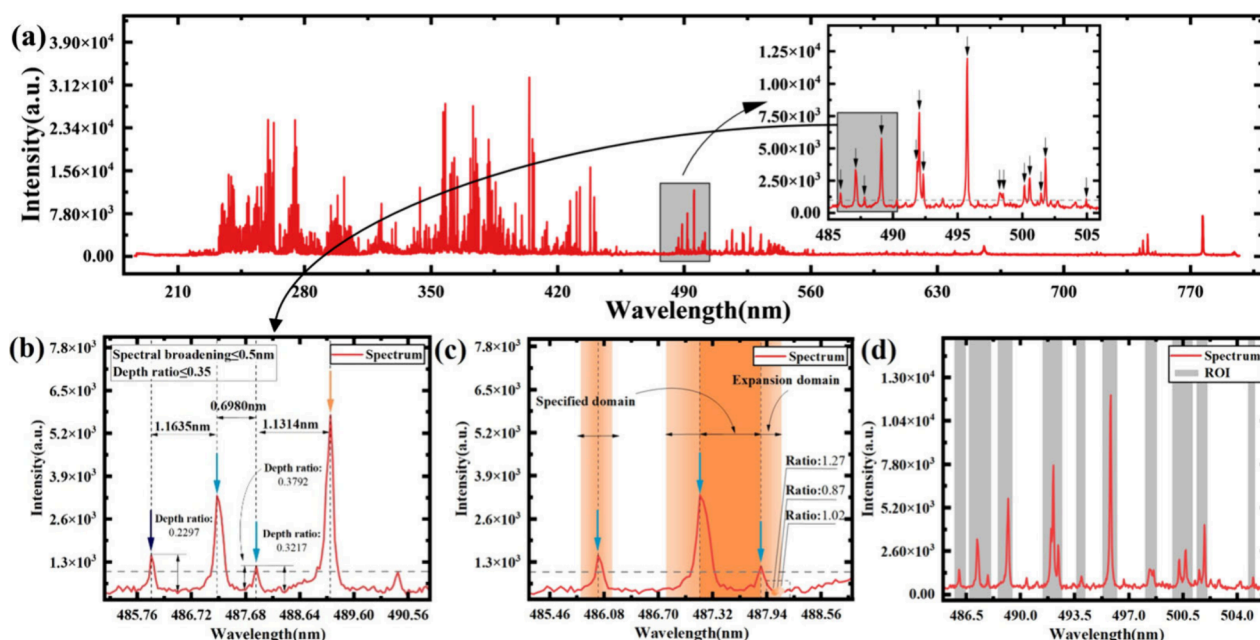


Figure 9. (a) Overview of full-spectrum ROI segmentation. Using the 485–506 nm range as a case study: (b) determination of interpeak distance and depth ratio, (c) determination of ROI boundaries (length), and (d) final ROI segmentation results.

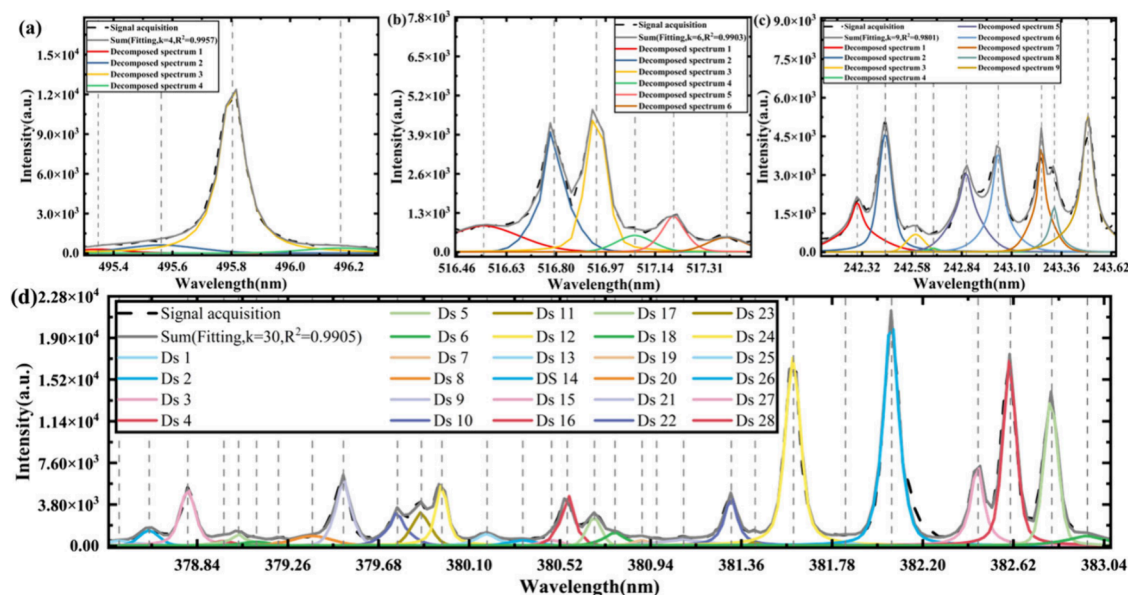


Figure 10. Decomposition of (a) a simple ROI, (b) a moderately complex ROI, (c) a complex ROI, and (d) a highly dense complex ROI.

between a "Voigt-shaped peak" and random noise. The hypotheses were defined as follows:

Null Hypothesis H_0 : The residual contains no omitted Voigt lines $y_i = C + \varepsilon_i$

Alternative Hypothesis H_1 : The residual contains a specific Voigt profile $y_i = C + A \cdot v_i + \varepsilon_i$.

A linear regression model with a fixed Voigt shape, $r(\lambda_i) = C + AV(\lambda_i) + \varepsilon_i$, was applied to the suspected peak region. A t test on the coefficient A was performed to determine the significance of the Voigt component. The result yielded $t = 14.31$ and $p = 1.4 \times 10^{-4} < 0.01$, indicating the presence of a statistically significant Voigt-shaped line. Cross-referencing with the NIST database identified this feature as the Fe I line at 313.25 nm. Following the incorporation of this line via the residual completion mechanism, the goodness of fit R^2 for the

interval improved from 0.9850 to 0.9935. Notably, the goodness of fit for the Ni I line increased substantially from 0.5300 to 0.8692, with its emission position accurately aligned.

Subsequently, the proposed method was applied to decompose the entire experimental spectrum, the distribution of which is shown in Figure 9(a). The full spectrum was segmented into 149 ROIs. To illustrate the ROI segmentation process without listing all regions, the 485–506 nm spectral range is selected as a representative case study.

First, the algorithm identifies the locations of characteristic spectral features. To distinguish valid signals from noise, spectral lines with intensities exceeding 1000 were selected. As shown in Figure 9(b), three emission positions were identified at 485.008, 487.172, and 487.870 nm, while the peak at

490.379 nm was excluded as it failed to meet the threshold requirement.

Next, the criteria for spectral aggregation are evaluated. Selected lines are grouped into a single cluster if they meet specific proximity or interaction conditions: (a) the separation distance is less than 0.5 nm; or (b) the depth ratio is more than 0.35.

Once the feature locations are established, the ROI boundaries are determined. To ensure sufficient spectral redundancy, the algorithm allows for a rapid intensity decay phase. Specifically, if the intensity difference between adjacent data points exceeds 30%, the spectrum is considered to be in a "rapid descent" phase. Conversely, if the variation across three consecutive points remains within 30%, the spectrum is deemed to have entered a "plateau" phase. As illustrated in Figure 9(c), the ratios of three consecutive points to their predecessors are 1.27, 0.87, and 1.07, respectively. These values indicate that the spectrum has stabilized, triggering the ROI cutoff and completing the data acquisition for that region. Consequently, the 485–506 nm range was segmented into 9 distinct ROIs, as shown in Figure 9(d).

Following ROI segmentation, the algorithm proceeds to solve for each region. The spectral regions are generally categorized into four types: simple ROIs containing a single peak (Figure 10(a)), moderately complex ROIs with few emission lines (Figure 10(b)), complex ROIs containing multiple lines (Figure 10(c)), and highly dense complex ROIs (Figure 10(d)). Analysis of the goodness of fit within these intervals indicates that the method achieves satisfactory results in practical spectral decomposition, with R^2 values reaching 0.9957, 0.9903, 0.9801, 0.9905, respectively.

Notably, as observed in Figure 10(a), although only one prominent emission feature is visually apparent within the ROI, the system extracted four decomposed components. This suggests that component generation in experimental spectra is more intricate than in synthetic simulations. Nevertheless, the total count of identified characteristic spectral lines increased from 634 to 1042. To validate the physical rationality of these components, the NIST Atomic Spectra Database was employed for cross-referencing. Additionally, Pearson correlation coefficients and Relative Standard Deviation (RSD) were introduced to evaluate the quantitative reliability of the decomposed components.

Validation and Quantitative Analysis

Upon obtaining the decomposed components, each was subjected to individual validation. Taking the four components from Figure 10(a) as an example (located at 495.347, 495.562, 495.804, and 496.171 nm), the evaluation results are presented in Figure 11.

First, the components were cross-referenced with the NIST Atomic Spectra Database. Only three matches were identified: Cr I at 495.371 nm, Fe I at 495.759 nm, and Fe I at 496.191 nm. Their corresponding NIST Accuracy (ACC) were D, B+, and E, respectively, with RSD values of 22.19%, 10.74%, and 16.27%. Generally, an RSD below 20% is required for quantitative usability, and a high ACC rating is essential for accurate spectral estimation. Consequently, the components with high RSD or low accuracy were discarded, and only the component at 495.804 nm (corresponding to Fe I at 495.759 nm) was retained for this interval. This single component yielded an R^2 of 0.9920 within the ROI. By applying this synchronous screening strategy to both the decomposed

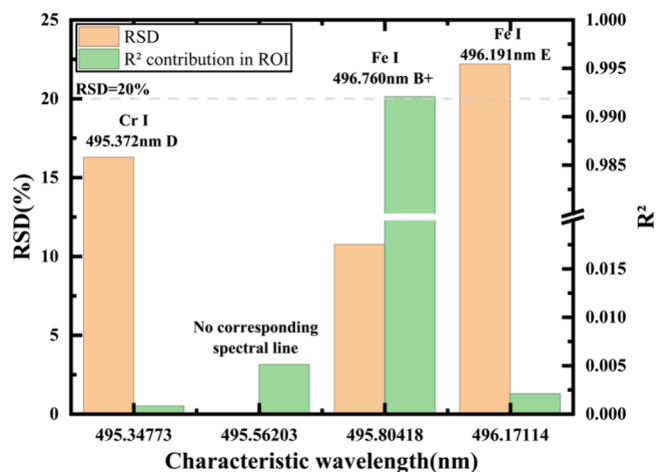


Figure 11. Performance evaluation of decomposed spectral lines.

features and the originally extracted features, the final counts of valid characteristic lines were refined to 705 and 630, respectively.

Following the verification of the mathematical reliability of the spectral lines, the next step involves evaluating the correlation between line intensity and concentration. In principle, LIBS spectral intensity exhibits a positive correlation with elemental concentration. Consequently, if the intensity fails to increase significantly with rising concentration, it suggests the presence of limiting factors such as saturation, self-absorption, or spectral overlap.

A mixed spectral feature was observed in the vicinity of 284.06 nm. As shown in Figure 12, decomposition revealed that this feature is primarily composed of two components, identified via database referencing as Fe II (284.064 nm) and Cr II (284.036 nm).

A comparison of the Fe II performance before and after decomposition shows that the R^2 value improved from 0.0157 to 0.2552. However, the enhancement was limited because the Fe concentration is extremely high (~95%), placing the line in a saturation regime where intensity does not scale linearly with concentration. In contrast, the recovery of the Cr II line was significant. This line remained undetectable via conventional peak-selection schemes but was successfully extracted through decomposition. Since Cr is present at lower concentrations, the line exhibits a high correlation between intensity and concentration, yielding an R^2 of 0.8576. Following a systematic verification of the remaining spectral lines using this approach, the number of valid features for the decomposed and original sets was reduced to 409 and 315, respectively.

To verify whether the decomposition process alters the physical consistency of the spectral signals, a plasma temperature validation was conducted to monitor any drastic fluctuations before and after decomposition. Twenty spectral lines exhibiting low interference were selected from the original spectra for Boltzmann plot analysis, as shown in Figure 13(a). The resulting slopes were -0.8506 , -0.8444 , -0.8150 , and -0.9098 . Using the Boltzmann slope formula $m = -1/(k_B \cdot T)$, where the Boltzmann constant $k_B \approx 8.61733326 \times 10^{-5}$ eV/K, the calculated plasma temperatures were 13642, 13742, 14238, and 12755 K, respectively. The average temperature was 13594 K, with correlation coefficients consistently exceeding 0.85.

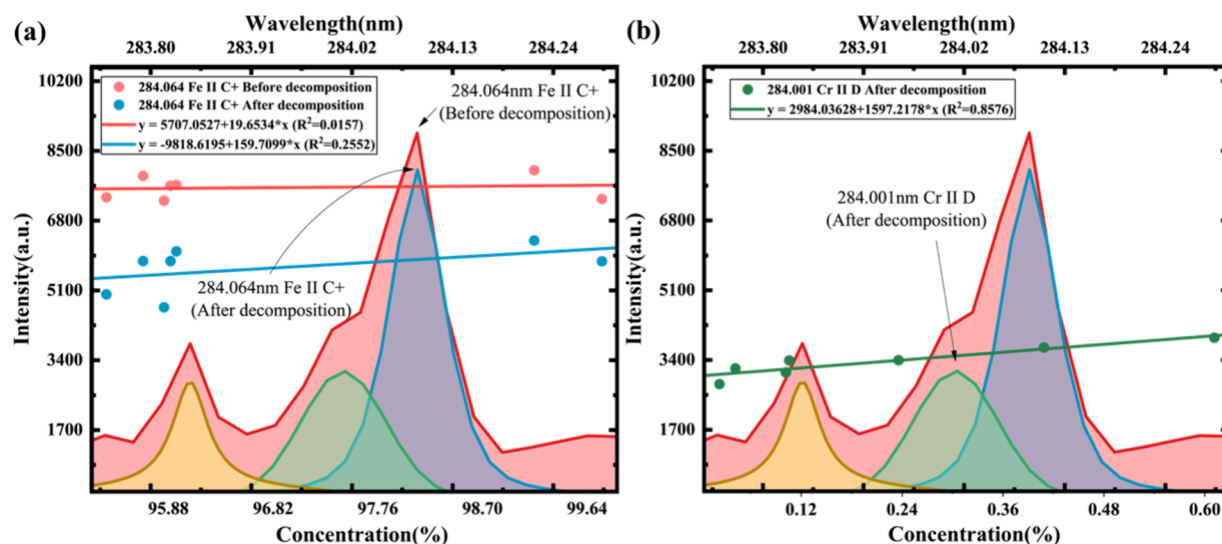


Figure 12. (a) Intensity–concentration relationship for Fe II at 284.06 nm before and after decomposition. (b) Intensity–concentration relationship for the decomposed overlapping Cr II feature at 284.036 nm.

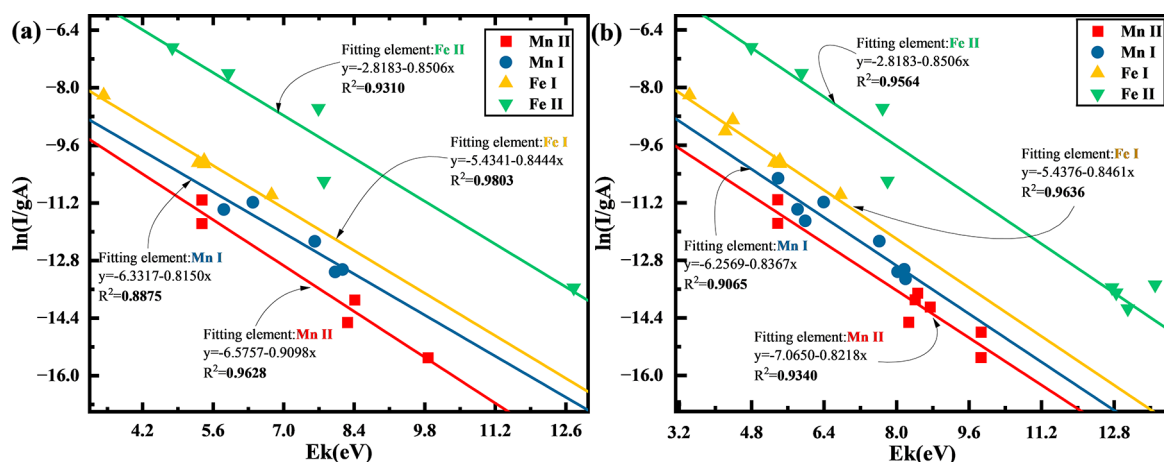


Figure 13. Plasma temperature validation: (a) before decomposition and (b) after decomposition.

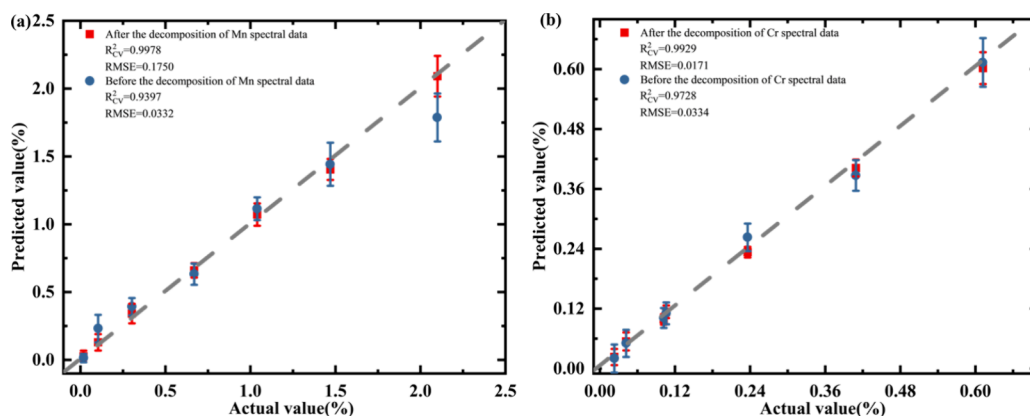


Figure 14. Quantitative analysis results before and after decomposition for (a) Mn and (b) Cr.

When the decomposed components were substituted into the calculation, the slopes shifted to -0.8506 , -0.8461 , -0.8376 , and -0.8218 . The corresponding plasma temperatures were recalculated as 13642, 13715, 13869, and 14120 K, yielding an average temperature of 13837 K. Notably, the correlation coefficients improved to a high level, exceeding 0.9.

The minimal shift in average temperature demonstrates that the decomposition preserves the plasma's physical characteristics, while the increased correlation indicates an improvement in the precision of the temperature calculation.

Finally, to assess the enhancement in quantitative performance following decomposition, two minor elements, Mn and

Cr, were selected for analysis. Fifteen characteristic spectral features were extracted for each element. Given the limited data set consisting of only seven concentration gradients, Leave-One-Out Cross-Validation (LOOCV) was employed to validate the data set. The calibration models were constructed using PLS, widely regarded as the standard regression method in LIBS analysis.

The quantitative results are illustrated in Figure 14. As shown in Figure 14(a), the R^2 for Mn improved from 0.9397 to 0.9978 after decomposition, while the Root Mean Square Error (RMSE) decreased significantly from 0.1750 to 0.0332. Similarly, Figure 14(b) indicates that the R^2 for Cr increased from 0.9728 to 0.9929. Although the improvement in linearity was modest, the RMSE exhibited a substantial reduction from 0.0334 to 0.0171, demonstrating enhanced predictive accuracy.

In this study, we proposed a spectral decomposition framework tailored for LIBS, explicitly addressing the characteristics of spectral interference. The method proceeds by segmenting the spectrum into ROIs to identify initial decomposition anchors. A residual inspection mechanism is then employed to recover missed features, while iterative parameter tuning ensures an optimized goodness of fit. The reliability of the extracted spectral lines was rigorously validated through NIST database cross-referencing, correlation analysis, and plasma temperature consistency checks. Ultimately, PLS modeling demonstrated that the proposed decomposition strategy significantly enhances quantitative performance.

A limitation of this study is that experimental validation is performed on Fe-rich steels. Although the proposed ROI partitioning is primarily resolution-normalized (for example, the clustering threshold scales with fwhm and the ROI width is constrained) and does not assume a specific matrix element, different matrices may exhibit more extreme strong–weak overlap patterns. Future work will therefore extend the evaluation to nonferrous matrices such as Al- and Cu-based alloys and controlled strong–weak overlap cases and will report quantitative comparisons of ROI stability and fitting performance across matrices. A remaining challenge arises from subresolution overlaps that are below the instrumental resolving limit of approximately 0.03 nm. In this regime, multiple transitions may form an unresolved mixed profile, and the decomposition should be interpreted as an effective mixed feature rather than a one-to-one separation of individual lines. To avoid propagating such nonidentifiable mixed features into quantitative analysis, the proposed framework incorporates a correlation-based feature screening step. ROIs dominated by strong mixing typically exhibit reduced correlation with any single target concentration when other constituents vary across samples; consequently, these ROIs fail the correlation criterion and are excluded from the final feature set used for calibration. This design choice provides a practical safeguard against extreme overlaps under the instrumental resolution constraint and will be further refined in future work.

CONCLUSION

This study proposes a spectral decomposition framework tailored for Laser-Induced Breakdown Spectroscopy signals, which optimizes decomposition performance primarily through ROI segmentation, decomposition anchor determination, and the introduction of a residual completion mechanism. Through the stepwise decomposition and fitting of spectral data, complex signals can be effectively disentangled

to recover the intrinsic information on individual spectral lines, thereby enhancing the accuracy of quantitative analysis. The effectiveness of the proposed method was validated through the decomposition and analysis of spectral data from steel alloy samples with varying concentrations. Experimental results indicate that the decomposition point identification mechanism successfully locates the majority of emission positions, achieving a goodness of fit exceeding 0.95 within intervals of the simulated data set. Furthermore, the residual completion mechanism effectively retrieves components from omitted data, raising the goodness of fit for individual components to above 0.9. In the processing of actual experimental spectra, the combined action of these two mechanisms ensures that the goodness of fit for the majority of spectral regions exceeds 0.95. Simultaneously, the decomposition process facilitates the identification of components exhibiting high correlation with elemental concentrations. The decomposed spectral lines not only align highly with true spectral positions but also demonstrate significant improvements in the quantitative analysis of plasma temperature and elemental concentration. Specifically, while maintaining consistent temperature trends, the minimum goodness of fit for the temperature calculation improved from 0.8875 to 0.9065. Finally, in the quantitative analysis, the R^2 for Mn and Cr increased from 0.9397 and 0.9728 to 0.9978 and 0.9929, respectively, while their RMSE values decreased to 0.0332 and 0.0171. These results confirm the validity of the proposed decomposition scheme. Although the current method has achieved demonstrable success in experiments, challenges regarding computational efficiency and fitting stability remain. Future work will focus on integrating advanced machine learning and deep learning techniques to optimize the algorithm, realizing more efficient and automated spectral decomposition. With these optimizations, the accuracy of quantitative analysis is expected to improve further, particularly for high-complexity and high-density spectral scenarios.

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Notes

The authors declare no competing financial interest.

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